REENGINEERING CORE BUSINESS PROCESSES THROUGH AN INTEGRATED IS-BASED SOLUTION FOR A MULTI-BRANCH RETAIL ENTERPRISE

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter provides background on the study by describing the organization's current situation and the information-related problems it is experiencing. It explains how information is currently managed, identifies issues affecting coordination and decision-making, and highlights the need for improvement. Lastly, this chapter introduces the proposed information system and explains how it can help improve information management and support better organizational decisions.

Introduction

Businesses and organizations rely on information systems to support daily operations and facilitate interactions with customers, suppliers, and other stakeholders. In modern enterprises, information systems are no longer limited to automating tasks; they serve as essential components for processing information, coordinating activities, and supporting managerial decision-making. Many organizations are structured around information systems that manage complex information flows and interrelated business processes (Mulaydinov, 2024). As information-based enterprises, organizations treat data and information as critical resources for achieving operational efficiency and strategic objectives (Pearlson et al., 2024).

Despite advances in digital technologies, many small and medium-sized enterprises continue to experience challenges in managing information effectively. Fragmented information environments, manual documentation practices, and isolated

systems often result in inconsistent records, delayed reporting, and limited visibility of operations. These issues are not merely technological limitations but reflect weaknesses in information governance and process coordination. When information is stored in silos and managed independently by different units, organizations struggle to maintain data accuracy, ensure traceability, and produce reliable consolidated reports.

In multi-branch retail enterprises, the need for coordinated information management is particularly critical. Core business processes such as sales transactions, inventory management, procurement documentation, billing, and reporting generate large volumes of data across different branches. Without an integrated information system, each branch may maintain separate records, leading to redundant data entry, inconsistencies, and delayed consolidation of reports. As transaction volumes increase, reliance on manual coordination and informal communication can limit managerial oversight and reduce operational efficiency.

From an information systems perspective, organizational effectiveness depends on the alignment between business processes and information flows. When information systems are fragmented, process coordination becomes dependent on manual intervention, increasing the risk of errors and limiting visibility across organizational units. Therefore, redesigning information workflows through an integrated system becomes essential to improve data reliability and support managerial oversight.

To address these challenges, this study proposes the development of an integrated transaction processing and information coordination system designed to improve the management, visibility, and consistency of organizational information across multiple branches. The proposed system aims to support decision-making by providing accurate,

timely, and structured information while strengthening transparency and accountability within the enterprise. Through upgrading manual processes using technological innovation and enhancing the enterprise's capacity for growth, the study aligns with the principles of the United Nations Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth. This goal emphasizes achieving higher levels of economic productivity through diversification, technological upgrading, and innovation, specifically supporting the formalization and growth of micro-, small-, and medium-sized enterprises (MSMEs).

Project Context

Gerlyn Variety Store operates as a retail and wholesale business with multiple branches in Baliwag, Bulacan. As the business expanded, its information processes-covering inventory management, sales transactions, customer orders, purchasing, employee administration, and supplier records remained managed independently at the branch level. Although point-of-sale (POS) systems are used at store counters to record sales transactions, they operate in isolation and are not integrated across branches or with other internal processes. Thus, the organization relies on a combination of manual records and basic digital tools to manage critical information.

Under the current information environment, business data are created, stored, and processed separately by each branch. Sales records, inventory levels, customer orders, and purchasing information are maintained independently, while data consolidation across branches is performed manually by branch managers. Stock deliveries from suppliers are documented manually. These practices result in fragmented information

storage, limited coordination, and the absence of a centralized repository for organizational data.

The existing setup presents several limitations related to information management. The lack of system integration leads to redundant data entry, inconsistent records, and delays in generating consolidated reports. Management has limited visibility into real-time inventory movement, sales performance, and wholesale transactions across branches, making monitoring and analysis time-consuming and prone to error. Manual handling of employee records, inventory receipts, and delivery documentation further constrains the availability of accurate and up-to-date information. Moreover, the absence of standardized information workflows and digital audit trails weakens information governance and makes transaction tracing and verification difficult.

These conditions hinder the organization's ability to produce reliable reports, support timely managerial decision-making, and effectively oversee operations across multiple branches. As transaction volumes increase and operational complexity grows, the limitations of the current information environment become more pronounced, affecting efficiency, coordination, and control within the enterprise.

Purpose and Description

The purpose of this study is to reengineer the core business processes of Gerlyn Variety Store across its multiple branches through an integrated information systems based approach. The study addresses the challenges arising from fragmented information management and limited coordination among branches, which have constrained operational efficiency and managerial decision-making as the organization expanded.

These conditions have reduced information visibility, delayed reporting, and increased the risk of inconsistent and unreliable data.

This study focuses on the role of an integrated information system in improving the coordination, accessibility, and quality of organizational information. Through establishing a centralized and shared information environment, the proposed approach seeks to support consistent data handling and structured information flows across branches. Centralized visibility of business information enables management to obtain timely and accurate insights into operations, facilitating improved monitoring and control without altering the organization's core decision-making responsibilities.

In addition, the study emphasizes the importance of information governance and accountability in a multi-branch retail enterprise. Standardized information processes and consolidated records are expected to strengthen data reliability, traceability, and reporting consistency. Improved governance supports clearer accountability for transactions and enhances management's ability to verify information and evaluate operational performance across branches.

This study aims to support Gerlyn Variety Store's operational sustainability and continued growth by showing how an integrated information system solution can serve as a strategic enabler for effective planning, monitoring, and decision-making in a multi-branch retail enterprise.

General Objective

The general objective of this study is to design and evaluate an integrated information system solution based on process analysis to improve the management,

coordination, and utilization of information related to sales and order processing, inventory information flow, procurement documentation, billing records, and reports in Gerlyn Variety Store, thereby supporting effective decision-making, improving information accuracy, and strengthening organizational efficiency.

Specific Objective

Specifically, this study aims to achieve the following objectives:

1. To analyze the current information process used by Gerlyn Variety Store and identify information-related issues affecting coordination, accuracy, and decision-making.
2. To design and develop an information system that improves the collection, organization, and flow of information related to the Sales & Order, Inventory, Procurement, Billing, and Reporting processes within Gerlyn Variety Store.
3. To evaluate the proposed information system solution based on:

    3.1. Business Process Reengineering Framework

        3.1.1. Process Efficiency Improvement

        3.1.2. Process Effectiveness

        3.1.3. Workflow Automation Impact

        3.1.4. Process Integration & Coordination

        3.1.5. Service Delivery Improvement

        3.1.6. Process Control & Monitoring

        3.1.7. Innovation & Value Creation

        3.1.8. Overall Process Improvement Impact

3.2. ISO 25010 System and Software Quality Standard

3.2.1. Functional Suitability

3.2.2. Performance Efficiency

3.2.3. Compatibility

3.2.4. Usability

3.2.5. Reliability

3.2.6. Security

3.2.7. Maintainability

3.2.8. Portability

Scope and Limitations

This study focuses on the design and development of an integrated information system to support the core business processes of Gerlyn Variety Store across its multiple branches in Baliwag, Bulacan.

The project covers information flows related to:

- Sales and order processing
- Inventory management and stock movement tracking
- Procurement documentation and supplier transactions
- Billing records
- Management reporting for retail and wholesale operations

The system will provide a centralized information environment that consolidates data generated by branch-level operations. It will support structured recording,

processing, retrieval, and reporting of product information, customer orders, purchasing records, and sales transactions.

The system will implement role-based access control for administrators, branch managers, store staff, suppliers, and customers to ensure secure and appropriate access to information. Standardized workflows will be implemented to promote consistency, traceability, and accountability across branches.

Limitations of the Study

The proposed system is subject to the following limitations:

1. **No online payment processing or financial system integration.** The system will not process financial transactions directly through electronic payment gateways. While users can select their preferred payment method within the system, the actual transfer of funds occurs through external manual processes. The system's role is limited to recording payment confirmation (e.g., uploading a reference number or marking an order as 'Paid' by an admin) rather than executing the financial exchange.
2. **No delivery and logistics management integration.** The system will not include automated delivery scheduling, courier booking, or real-time shipment tracking through third-party logistics providers.
3. **Limited human resource management coverage.** The system will not include comprehensive human resource management functions such as recruitment, attendance tracking, performance evaluation, payroll disbursement, or statutory compliance processing. While the system records

staff identifiers for audit trails and basic accountability logs, it does not perform payroll processing, performance evaluations, or attendance tracking.

4. **No integration with external enterprise systems.** The system will operate as a standalone internal information system and will not integrate with external accounting software, supplier systems, or third-party platforms.
5. **Exclusion of Point-of-Sale (POS) Hardware and Software:** The study will not replace the existing POS system currently utilized by Gerlyn Variety Store. The proposed system is designed to function as a back-end integrated information system that supplements the existing POS by handling higher-level inventory, procurement, and multi-branch consolidation. Data from the existing POS will be manually or periodically encoded into the proposed system for centralized reporting.

These limitations are established to maintain focus on improving core business information flows and ensuring the feasibility of system development within the scope of the study.

Definition of Terms

The following terms are operationally defined to provide a clearer understanding of the specific methodologies, frameworks, and technical concepts utilized within the context of this study

Agile Methodology - The iterative development approach to be used in this study wherein the proposed system for Gerlyn Variety Store will be built and refined through

iterative cycles, incorporating feedback from the store owner and staff during each phase to ensure that the system aligns with actual business processes.

Billing Records - The system-generated digital records that will be used in this study to document and monitor all sales transactions, including wholesale and retail invoices, enabling the store to track customer payments and outstanding balances.

Business Process Reengineering - The strategic framework used to systematically redesign Gerlyn Variety Store's existing manual processes such as sales recording, inventory tracking, and procurement into streamlined, system-supported workflows to improve efficiency, coordination, and data accuracy.

Centralized Data Management - The system's implementation of a single database where all branch transactions of Gerlyn Variety Store, inventory updates, and billing records will be stored and accessed, ensuring consistency of information across the entire store operation.

Decision Support - The system's capability to generate timely and accurate information such as sales summaries, stock levels, and transaction reports that will be used by Gerlyn Variety Store's owner and manager to guide operational and strategic decisions.

Information Governance - The set of rules, roles, and processes implemented within the system to ensure that the store's data is accurate, secure, and handled consistently across all branches.

Integrated Information System - The proposed system developed in this study that will connect key business functions such as sales, inventory, and procurement

allowing data to flow automatically between processes without duplication or manual re-entry.

ISO 25010 - The international standard for systems and software quality used as a benchmark in this Gerlyn Variety Store to evaluate the proposed system's functional suitability, reliability, and usability.

Kanban - The task management technique used by the researchers during the development of Gerlyn Variety Store's system development to organize, visualize, and monitor progress of development activities through categorized workflow stages.

Management Reporting- The generation of consolidated reports from system data to support monitoring, evaluation, and decision-making by the branch manager and owner.

Procurement - The digital process of recording Gerlyn Variety Store supplier transactions, purchase orders, and stock receipts within the system to ensure a clear audit trail of incoming goods.

Real-Time Information - The immediate availability of updated data within the system that provides the owner with an instant view of current stock levels and sales activities across all locations.

Role-Based Access Control (RBAC) - The security configuration within the system that restricts or grants access to specific features based on the user's position from Gerlyn Variety Store as Admin, Manager, Staff, Customer, or Supplier.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents a review of related literature and existing systems relevant to the proposed study. It discusses various concepts, theories, and prior studies related to information systems, business process reengineering, information governance, and decision support. The chapter also examines existing manual and computerized systems that perform similar functions, identifying their strengths and limitations. This review highlights gaps and opportunities for improvement to justify the development of the proposed information system. Lastly, this chapter presents the study's conceptual framework, which illustrates how information flows through the system and supports organizational processes and decision-making.

Related Literature

Reviewing related literature gives an understanding of how information systems help organizational operations, how information flow affects coordination and decision-making, and how integrated systems improve efficiency. This literature explains the importance of well-managed information, especially in organizations with multiple branches and units, such as retail enterprises.

Business Process Analysis and Mapping

Business process analysis is the systematic examination of an organization's current workflows to identify inefficiencies, redundancies, and bottlenecks. This phase is

critical because it provides the factual foundation for any subsequent system development. Researchers can pinpoint where manual data entry leads to errors or where information flow is interrupted between branches through analyzing the current business process. According to Li and Nazif (2022), process analysis allows organizations to move beyond simple task removal by identifying specific structural bottlenecks and improving cross-departmental coordination.

A primary tool in this phase is process mapping, which provides a visual representation of how information moves between different units. As stated by Wang et. al., (2024), combining process reengineering with rigorous workflow mapping is the most effective way to significantly reduce operational waste.

Business Process Reengineering

Business Process Reengineering (BPR) is a strategic approach that involves fundamental rethinking and radical redesign of organizational workflows to achieve significant performance improvements. BPR restructures processes in order to align with organizational goals and requirements. Fetais et al. (2022) emphasize that BPR targets specific areas within key performance indicators such as cost, quality, service, and speed by eliminating redundant tasks and streamlining workflows.

Li and Nazif (2022) explained that BPR helps organizations to improve workflows by identifying bottlenecks and improving cross-departmental coordination beyond simple task removal. This is supported by Wang et al. (2024), who also explained that combining process reengineering with workflow analysis and mapping significantly reduces operational waste. These findings suggest that a systematic, top-down redesign of

processes is important for enhancing productivity and ensuring the efficient use of organizational resources in a competitive environment.

Organizational Information Flow and Integration Challenges

The transition from manual documentation to integrated digital systems is a significant factor in the operational sustainability of the Philippines' small and medium enterprises (SMEs). According to a study by Gabriel (2025), most local retailers believe that manual record-keeping is a sufficient standard for daily operations. However, these beliefs often lead to "moderately pressing problems" such as losing track of items and having incorrect data. To understand this further, the concept of inventory management must be studied. Inventory management is the process by which a business tracks the products it has in stock to meet customer demand (Gabriel, 2025). When records are kept only on paper, it becomes difficult for owners to see the status of different branches. This leads to data fragmentation, where information is stuck in handwritten logs and cannot be used for quick reporting.

Connecting different parts of a business is a major factor in growth and in staying competitive in the local market. According to Borazon & Supangco (2020), internal integration is the smooth connection between the sales, stock, and buying departments. The study shows that this connection is the strongest reason why small Philippine businesses succeed. To achieve this, companies use a Management Information System (MIS). An MIS is an organized tool that helps a business process data into useful information (De Castro et al., 2024). Instead of relying on slow, manual paperwork, an integrated system helps managers make better decisions by providing real-time sales data

(Lastimado et al., 2021). This makes the business more efficient by eliminating manual steps that waste time.

In a study by Osio et al. (2026), having a digital system is necessary for a business to survive disruptions. This is known as digital resilience. Digital resilience is the capacity of a store to recover quickly from disruptions by adopting technology. The study found that "digitally-capable" stores recovered much faster because they used systems to protect their finances and meet customer demand. In addition, it was emphasized that "organizational capability," or how well staff use the system, is more important than a large budget (Osio et al., 2026). These local studies show that moving from paper to a digital system is an important strategy for any business seeking transparency, organization, and long-term success.

Information Systems in Organizations and Decision Support

Information Systems in Organizations and Decision Support describes how organizations use information systems not only to record transactions, but also to provide information that supports managerial monitoring and decision-making. DSS are established information systems designed to improve decision-making through data and analysis. Phillips-Wren et al. (2021) explain that decision support system capabilities include retrieving information, enabling ad hoc analysis, producing aggregated reports, and helping decision makers estimate the effects of different choices. They also argue that business intelligence and analytics can be viewed as part of decision support because they aim to convert more data into deeper insights for decision-making. This supports the

study's goal of providing managers with clearer visibility across branches through consolidated, structured reports that guide operational and planning decisions.

However, many small and medium enterprises often struggle to fully leverage data for data-driven decision-making. Žilka et al. (2024) explain that digital technologies enable the rapid collection of large volumes of data, but SMEs often lack systems or solutions to use data effectively in managerial decision-making. As a result, managers may rely on manual consolidation or incomplete information. This situation is common in multi-branch retail businesses, where operational data are recorded separately at each branch, making it difficult to obtain a clear, consolidated view of business performance. These studies highlight the importance of integrated information systems that organize operational data and provide meaningful reports for managerial use.

Information Governance and Data Quality

Information governance ensures that organizational data is accurate, consistent, and properly managed. Zorrilla et al. (2025) explain that data governance helps organizations maintain reliable records by establishing policies, roles, and procedures for managing information. This improves data quality and supports decision-making. Proper governance also guarantees accountability and control over information access. Skrodelis et al. (2020) explain that governance frameworks help organizations define user roles, assign responsibilities, and manage access to information systems. These controls improve data security and ensure that information remains reliable.

Additionally, information quality is an important factor in the success of information systems. Al-Kofahi et al. (2020) explain that information systems must give

accurate, reliable, and accessible information to support users and organizational objectives. These findings emphasize the importance of structured data management, especially in integrated systems that consolidate information from multiple branches. In multi-branch retail and wholesale businesses, governance mechanisms ensure that operational data remains accurate and usable for reporting and monitoring. This is significant for coordinating workflows and making informed decisions.

Retail and Inventory Information Systems

Retail information systems focus on supporting the recording, monitoring, and management of operational data, such as sales transactions, inventory records, and purchasing activities. Integrated retail systems improve organizational visibility by providing accurate and centralized information. Studies have shown that applying information systems to retail operations improves efficiency and reduces errors. For example, Nargesi and Bazaee (2023) found that implementing integrated systems in warehouse and sales operations improved efficiency by reducing delays and improving coordination. Automation and centralized data management improved operational performance and reduced manual work.

These findings emphasize the importance of integrated systems in retail organizations, especially those operating across multiple locations. Through providing centralized access to operational data, integrated retail information systems help managers monitor inventory, track sales, and coordinate business activities more effectively.

Related Systems

Selected existing retail systems are presented to examine how current technologies support business operations such as sales, inventory, procurement, and reporting. Each system is analyzed based on its key functions, operational coverage, and limitations. The review aims to identify features that can be adapted and gaps that need to be addressed in the development of the proposed system. Through this comparison, the study establishes a basis for designing a more integrated and organization-specific solution that responds to the operational needs of the business.

Lightspeed Retail POS

Lightspeed Retail POS is a retail management system used by retail businesses to manage sales transactions and inventory operations. It is commonly used in stores that require efficient transaction processing and inventory monitoring. The system handles information such as product details, sales transactions, inventory levels, and basic sales reports. It uses a centralized database to record and organize operational data.

The strength of Lightspeed Retail POS is its ability to improve transaction recording and maintain organized sales and inventory records. It also supports reporting features that help monitor sales performance.

Although, the system primarily focuses on sales and inventory management. It has limited support for other operational processes such as procurement coordination and internal document tracking, which are also important in managing overall business operations.

The study will adapt the features of this system which demonstrates the importance of centralized transaction recording and inventory monitoring. Extending support to include procurement coordination and broader operational processes.

Odoo

Odoo is an integrated enterprise management system used by organizations to manage multiple business functions such as sales, inventory, purchasing, and reporting. It is widely used across various industries due to its all-in-one platform approach. The system handles a wide range of information including sales data, inventory records, purchasing transactions, supplier information, and operational reports within a centralized system.

Odoo's strength is its ability to integrate multiple business processes, allowing better coordination and improved accessibility of information across departments.

Although, the system is designed for general business use and may include features that are not necessary for smaller retail enterprises. This can make the system more complex to implement and use for organizations with simpler workflows.

This study will adopt the concepts of this system that shows the value of integrating multiple business functions into a single platform. The proposed project will focus on tailoring the features specifically to the operational needs of the organization to avoid unnecessary complexity.

Square Point of Sale

Square Point of Sale is a retail system used by businesses to manage sales transactions and inventory tracking. It is commonly used in small to medium-sized retail environments. The system handles information such as product sales, transaction records,

inventory updates, and sales reports. It automatically updates stock levels based on recorded transactions.

Its strength is in improving transaction efficiency and reducing manual errors through automated inventory updates and organized sales recording.

The system mainly focuses on sales and inventory management. It has limited support for other operational processes such as procurement coordination and consolidated reporting across multiple branches. As a result, businesses may still depend on separate systems for other functions.

This study will adopt the features of this system that demonstrates the effectiveness of automated inventory updates and accurate transaction recording. These features will be integrated into the proposed system while expanding functionality to support procurement processes and multi-branch reporting.

Synthesis of Related Literature and Systems

The reviewed literature and systems demonstrates that while digital tools for retail are common, the primary challenge for multi-branch businesses is integrating separate tasks into a single, unified workflow. The connection between these studies and the proposed research lies in the critical role of Business Process Reengineering (BPR). The literature proves that analyzing and redesigning current workflows is essential before building a system to ensure the technology actually improves speed rather than just digitizing old inefficiencies. While popular systems like Lightspeed and Square are effective for basic sales, they often lack the deep integration and data rules needed to link procurement, inventory, and reporting across multiple branches in one place. Therefore,

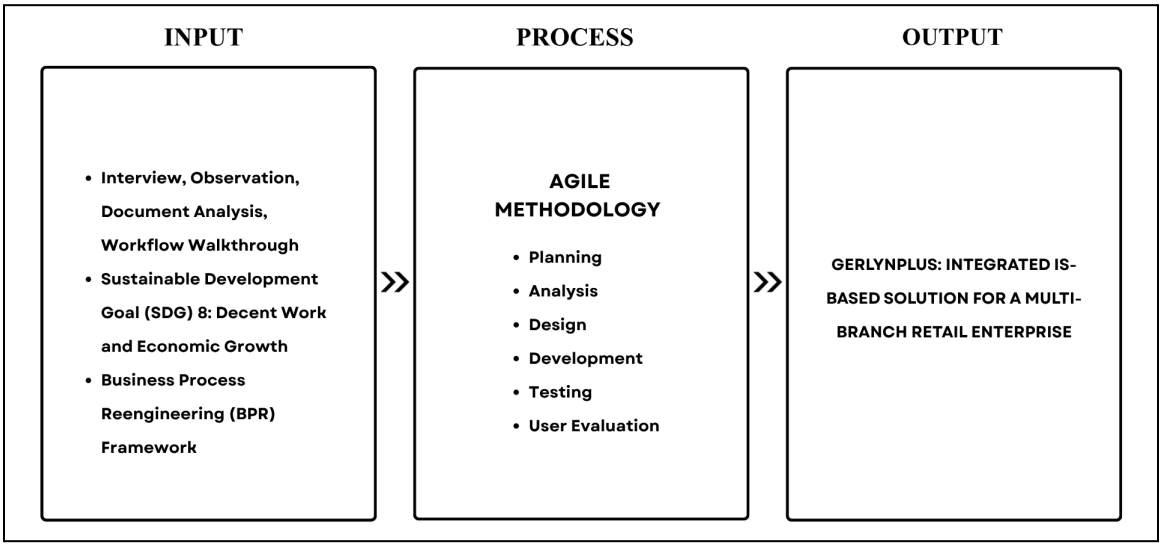
this study aims to fill these technical gaps by creating a system that goes beyond simple record-keeping to provide decision support to give managers the clear and accurate reports they need to make better business choices and maintain long-term digital resilience.

Conceptual Framework

The conceptual framework of the study presents the structure showing how the proposed system supports the transformation of core business process information into useful outputs. The framework uses the Input–Process–Output (IPO) model to illustrate the flow of information within the proposed integrated information system. The IPO model shows how operational data are captured as inputs, processed through system-supported business process workflows, and transformed into outputs that support organizational monitoring and decision-making. This conceptual paradigm is shown in Figure 1.

Figure 1

Conceptual Paradigm of the Study



The input component consists of important data and resources required for researcher's project development including product information, sales records, and inventory data from the client gathered through interview and document analysis. These inputs are integrated with Sustainable Development Goal (SDG) 8, which promotes sustained and inclusive economic growth by enhancing the operational efficiency of small-scale enterprises. This is coupled with the Business Process Reengineering (BPR) framework to identify manual inefficiencies and labor-intensive bottlenecks. This foundation serves as the basis for understanding current business processes and identifying specific areas for digital enhancement.

The process component represents the transformation phase governed by the Agile Methodology through iterative cycles of planning, designing, and testing. During this stage, manual workflows are systematically redesigned into digital processes and evaluated against ISO 25010 standards to ensure functional quality. The methodology

ensures that the development of the system undergoes constant refinement based on stakeholder feedback and organizational needs.

The output shows the successful reengineering of core business processes through the implementation of an integrated IS-based solution for a multi-branch retail enterprise. This results in the development of GerlynPLUS, a centralized IS-based solution designed to integrate sales, inventory, procurement, billing, and reporting processes. GerlynPLUS (Process-Linked Unified Systems Enterprise), which emphasizes the capability of the system to connect and unify core business processes into a single platform. This signifies the transition to a unified system that provides Gerlyn Variety Store with improved data accuracy and streamlined multi-branch coordination.

CHAPTER III

METHODOLOGY

This chapter presents the research design and systematic procedures that will be followed in analyzing, designing, and developing the proposed information system. It explains the methods that will be used to gather data, examine existing business processes, and identify system requirements. Furthermore, this chapter outlines the system development methodology, proposed system models, tools and technologies to be utilized, and the projected timeline for the completion of the study.

Research Design

This study will use a descriptive-developmental research design. The descriptive aspect will analyze the current workflow processes of Gerlyn Variety Store, particularly in sales, inventory management, procurement, billing, and reporting across multiple branches. Through this approach, the study will identify operational inefficiencies such as redundant data entry, fragmented record-keeping, delayed reporting, and weak inter-branch coordination. This allows the researchers to establish a clear understanding of the operational challenges encountered by the organization. Information systems play a significant role in supporting organizational operations and managerial decision-making by providing structured, reliable, and accessible information for monitoring and analysis (Gürkan, 2022).

On the other hand, the developmental aspect will be applied in the design and development of the proposed integrated information system solution based on the

identified needs. This approach ensures that the proposed system supports the creation of an improved workflow, system models, and a functional prototype aligned with the operational requirements of Gerlyn Variety Store. Integrated information systems improve operational efficiency by automating processes, reducing manual work, and providing consolidated information for better coordination and management oversight. The integration of descriptive and developmental approaches allows a structured and systematic process in both analyzing the current process and producing an effective solution.

This study uses a descriptive-developmental research design. Business Process Reengineering serves as the conceptual basis for workflow redesign, while the system is developed using the Systems Development Life Cycle (SDLC) framework with an Agile implementation approach. Kanban through Jira is used as a task management tool to organize development activities.

Business Process Reengineering Framework

The study adopts Business Process Reengineering (BPR) as the conceptual basis for improving business operations. The existing core processes: sales, inventory, procurement, billing, and reporting will be analyzed to identify inefficiencies such as delays, redundant encoding, fragmented records, and lack of coordination across branches.

The study follows a structured BPR approach by:

- Identifying existing core processes
- Diagnosing workflow inefficiencies

- Redesigning business processes
- Integrating information flows across branches
- Supporting the redesigned processes through a centralized information system

The redesigned workflows aim to eliminate duplication, reduce delays, and improve the accuracy and accessibility of information. These improvements will be supported by a centralized system that enables real-time data sharing and enhanced managerial oversight.

Data Gathering Methods

The researchers will employ four data gathering methods to capture the actual operational reality of the client and understand the current information management process.

According to Enago Academy (2023), research interviews play a pivotal role in collecting data for various academic, scientific, and professional endeavors. They provide researchers with an opportunity to delve deep into the thoughts, experiences, and perspectives of an individual, thus enabling a comprehensive understanding of complex phenomena. A structured formal interview will be used to identify the challenges of managing multiple branches and the specific limitations caused by current manual processes. This method allows the researchers to gather first-hand insights into the owner's operational pain points, coordination gaps, and the specific reporting needs required for high-level decision-making.

Observation, as defined by George (2023), is a research method where the characteristics or qualities of a phenomenon are described without the primary use of

quantitative measurements. Rather, the observation is based on the researcher's interpretation of the environment and activities. This will serve as a supporting method to validate the interview data. Direct observations at the store counters and administrative areas of the business will be carried out to pinpoint operational bottlenecks, such as redundant data entry tasks and manual coordination delays between shops. This method allows the researchers to witness the current process in real-time and identify delays, duplicate encoding, and operational efficiencies that may not be fully captured through verbal interviews alone.

Document analysis is a qualitative research technique that involves evaluating physical or digital documents to interpret their meaning and build upon the information they provide (Indeed Editorial Team, 2025). The researchers will perform a document analysis of specific physical records, including inventory lists of orders to suppliers and fragmented sales reports from different branches. This process aims to show the lack of digital audit trails and the risks associated with fragmented data storage. The researchers will formally request and obtain the owner's consent to review these internal documents, with the understanding that certain sensitive or highly confidential information may be withheld from viewing to maintain business privacy. This method identifies required data fields, forms, and missing audit trails.

A workflow walkthrough is a systematic procedure used to trace the path of a transaction or a piece of information through a business process. This will be executed to map the manual journey of business data from the initial sales transaction at the counter to the final generation of management reports. Through tracing these manual hand-offs step-by-step, the researchers can ensure that the proposed automated workflow is

grounded in actual requirements and addresses the specific "dead ends" in the current chain. Furthermore, this walkthrough ensures that the design of the new information flow remains strictly compliant with the Data Privacy Act of 2012 (RA 10173). This method identifies the actual sequence of activities and breakdowns in information flow. The gathered data will be used to define system requirements, redesign workflows, and develop the proposed system models and database structure.

Participants

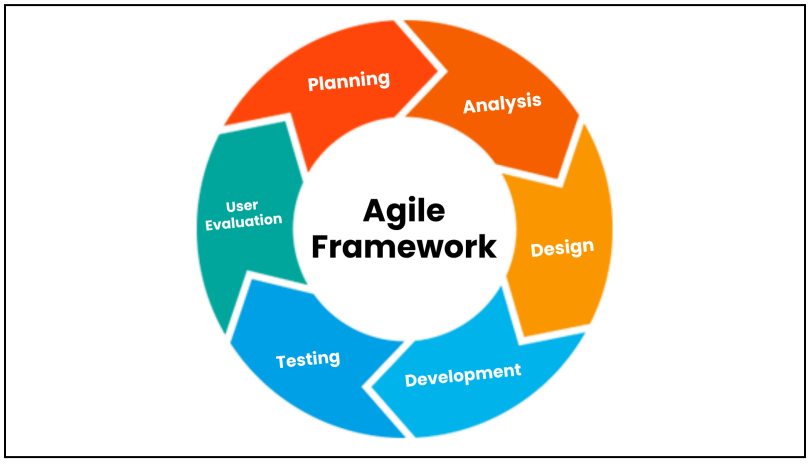
The evaluation of the proposed system will be conducted to selected participants to ensure both technical quality and user acceptability. The study will involve three (3) IT experts who will evaluate the system based on the ISO 25010 software quality standards, particularly focusing on criteria such as functionality, usability, reliability, efficiency, maintainability, and portability. Their assessment will provide a professional and objective validation of the system's technical performance and compliance with recognized quality benchmarks. Also, the user-side evaluation will be carried out to the intended end-users of the system, including the store owner and selected staff members from different branches. This group will assess the system in terms of usability and its effectiveness in supporting daily business operations. Their feedback is important in determining whether the system meets actual operational needs and improves the current workflow. These participants ensure that the system is both technically sound and practically applicable in the business environment.

System Development Methodology

The study will use an Agile methodology to guide the development of the project. Agile enables the project to be created in incremental cycles based on user feedback and evolving organizational needs. Activities are structured into key phases: planning, analysis, design, development, testing, and user evaluation. These phases ensure that the project is developed systematically while remaining flexible and responsive.The Development and Testing phases are carried out iteratively, allowing continuous improvement of system functionalities that support business operations and information management. SDLC provides a structured process that will guide development from planning up to evaluation, ensuring that the system meets user requirements and organizational needs (Jackson, 2024). Each iteration will focus on improving system support for business processes such as sales recording, inventory tracking, and reporting, ensuring better coordination and information management. This structure ensures that the system development remains aligned with the redesigned workflows produced through Business Process Reengineering.

Figure 2

Agile Framework



The development process will follow Agile iterations structured around key development activities such as planning, analysis, design, development, testing, and evaluation

Planning Phase

This phase will define the system objectives, scope, and key requirements based on the analysis of current workflows. It will focus on identifying how the system can support core business processes such as sales recording, inventory monitoring, procurement coordination, billing, and reporting. Emphasis will be placed on aligning system goals with the need for improved information flow and coordination.

Analysis Phase

This phase will involve examining existing business processes and identifying system requirements based on observed inefficiencies. A critical component of this phase is the conduct of a comprehensive risk assessment to identify potential technical, operational, and project management threats. The researchers will identify key risks including the project management risk or the timeline wherein the complexity of multi-branch integration may impact the delivery schedule; financial risk which concerns the budget for cloud services and hardware; and operational risk, regarding the staff's transition from manual to digital workflows. The analysis will also focus on understanding how information is currently generated, shared, and managed, and how these will be improved to support better coordination and decision-making.

Design Phase

The design phase will involve developing system models that will represent how information will flow within the proposed system. This stage will incorporate

benchmarking to ensure the system architecture meets industry standards for multi-branch retail synchronization and data integrity. It will be based on the redesigned workflows produced through the Business Process Reengineering (BPR) analysis to ensure that system functionalities directly support improved business processes. In addition, this phase will define the system's security architecture through the establishment of Role-Based Access Control (RBAC) which will map specific permissions to user roles, such as Owner, Manager, Staff, Supplier, and Customer to ensure data confidentiality and integrity. Tools such as Use Case Diagrams, Context Diagrams, Data Flow Diagrams, and Entity Relationship Diagrams will be used to ensure that the system structure supports integrated workflows and consistent data management.

Development Phase

The development phase will involve system functionalities implemented to support redesigned business processes. These include features that will facilitate sales transactions, inventory monitoring, procurement, and report generation. During this stage, the RBAC framework will be integrated into the system’s backend to enforce security boundaries and prevent unauthorized data modification. The focus will be on ensuring that these functionalities promote accurate data capture, efficient information flow, and coordination across branches.

Testing Phase

Testing will be conducted to ensure that the system supports business processes effectively and produces accurate and reliable outputs. This will include verifying that information will be properly recorded, processed, and reported. Issues identified will be addressed to improve system performance and usability.

User Evaluation Phase

The system will be evaluated by users based on the ISO/IEC 25010 quality model. Evaluation will focus on usability, functionality, reliability, and performance, as well as the system's ability to support workflow integration, information consistency, and managerial decision-making. Feedback gathered is used to further enhance the system. Evaluation will also assess how well the system improves information visibility, reporting timeliness, and inter-branch coordination.

Project Management and Workflow

The researchers will use the Kanban framework as a task management tool to support the Agile development activities. Kanban provides a structured way to organize tasks and monitor progress. It helps the team track work, manage workload, and improve workflow efficiency (Anderson & Carmichael, 2016).

The researchers will use Jira Software to implement this workflow. Jira helps organize tasks, monitor progress, and manage development activities (Atlassian, 2023). Kanban will manage development tasks throughout the SDLC process with Agile implementation. The researchers will create a digital board in Jira where tasks will appear as cards with details such as description, assigned member, priority level, and status. The board will include stages such as To Do, In Progress, Testing and Review, and Completed.

Tasks will move across stages based on progress. Work will begin in the To Do stage, proceed to In Progress, and move to Testing and Review for checking and feedback. Completed tasks will move to the final stage. Development activities such as

implementation of system components that support data capture, processing, and reporting using the Kanban board. Tasks will return to earlier stages when revisions are needed. This process will support continuous improvement and align with the Agile approach.

The use of Kanban in Jira will keep development activities organized, clear, and adaptable, supporting the successful completion of the proposed information system.

Figure 3

Project Management Tool (Jira)

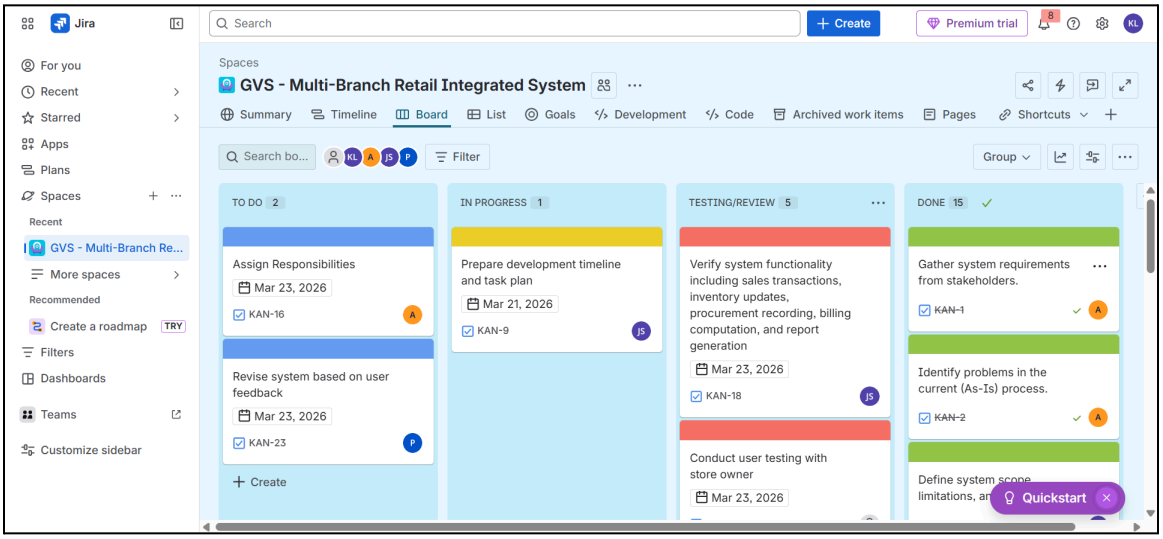
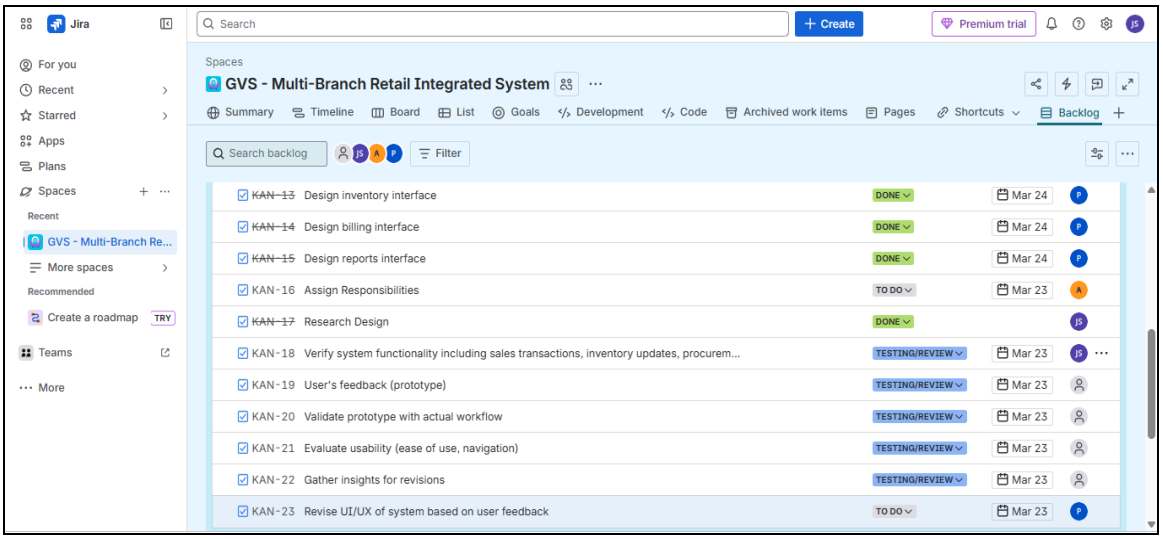


Figure 4

Jira-Based Kanban Task Management for System Development



Proposed System Models

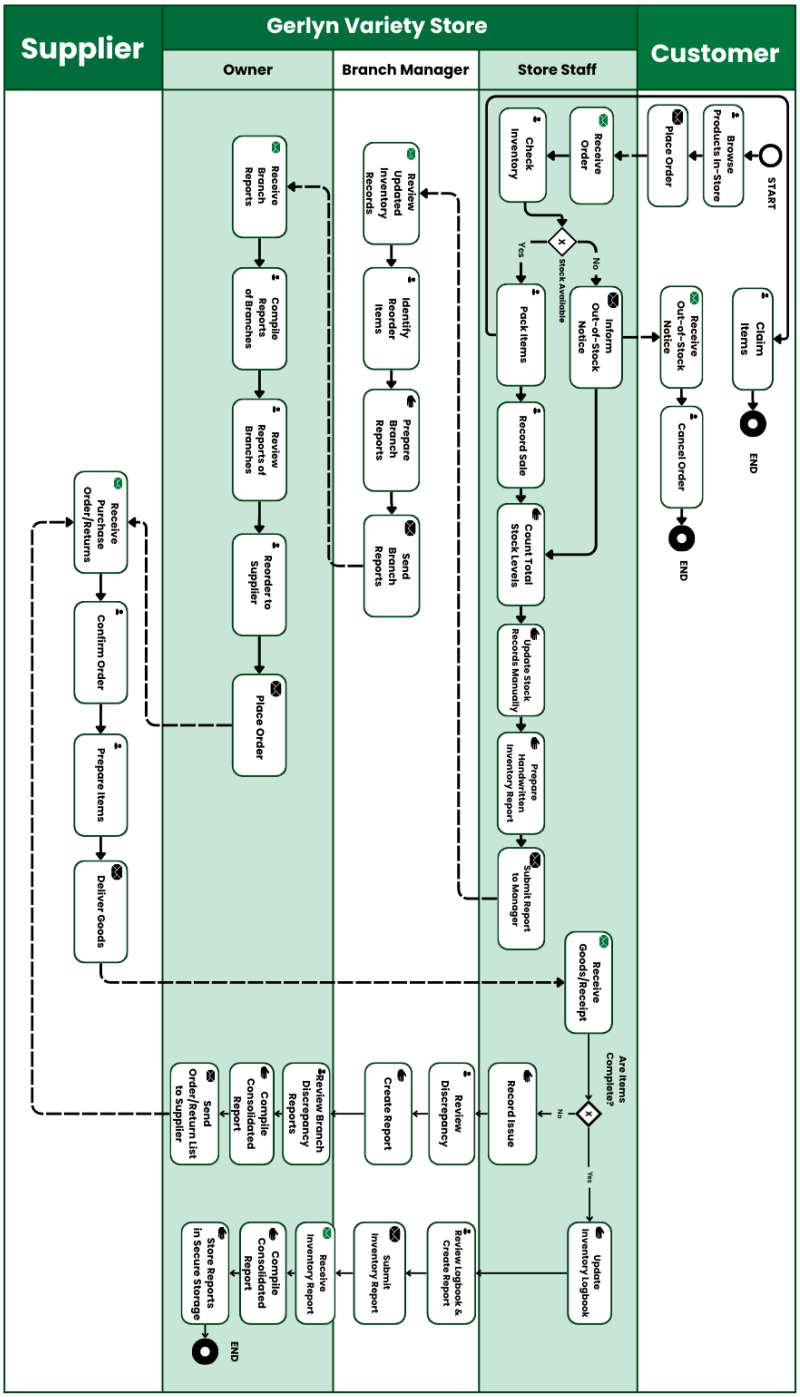
This section presents the proposed models that will guide the development of the system, including the current and proposed process models, Data Flow Diagram, and Entity Relationship Diagram.

As-Is Process Model

The As-Is process model illustrates the current workflow of Gerlyn Variety Store, which relies heavily on manual processes and decentralized coordination across branches.

Figure 5

As-Is Process Model



In the current process, customer orders are received through walk-ins or informal channels such as social media. Staff manually check product availability through physical inventory counts or logbooks. Sales transactions are recorded manually, and stock updates are not reflected in real time.

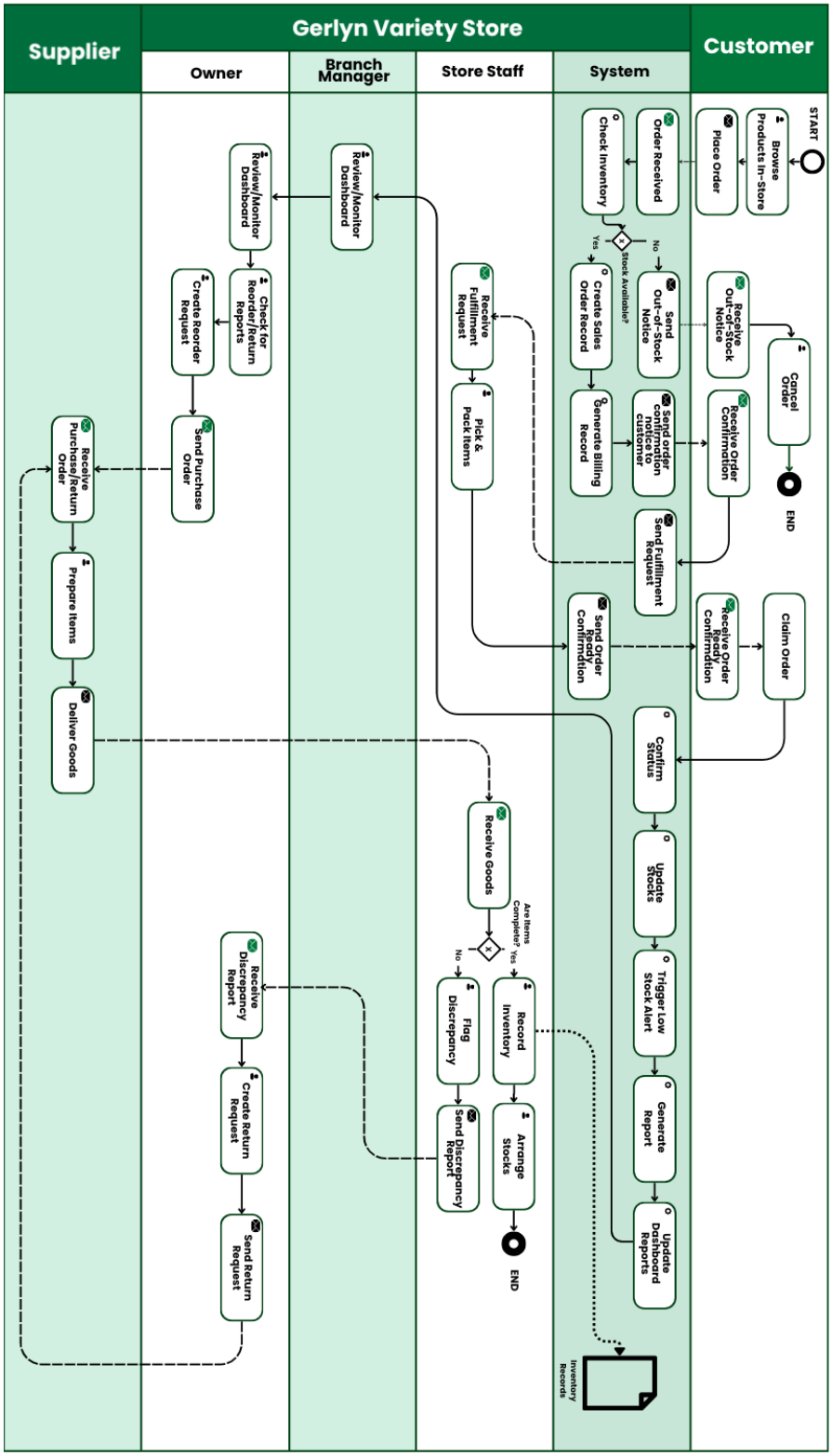
When items are out of stock, Staff coordinate verbally with the Branch Manager, who then prepares a restocking request. The request is forwarded to the Owner for approval before contacting the Supplier through calls or messaging applications. This process results in delays and potential inconsistencies in inventory records due to the absence of a centralized system.

To-Be Process Model

The To-Be process model presents the improved workflow using the proposed integrated information system, which introduces centralized data management and automation of key processes.

Figure 6

To-Be Process Model



In the proposed process, customer orders are directly entered into the system, where product availability is automatically checked against a centralized database. Staff can immediately process confirmed orders without manual verification.

The system continuously monitors inventory levels and automatically generates alerts when stock reaches the reorder point. The Branch Manager reviews these alerts and submits procurement requests to the Owner for approval. Once approved, purchase orders are sent to the Supplier through the system.

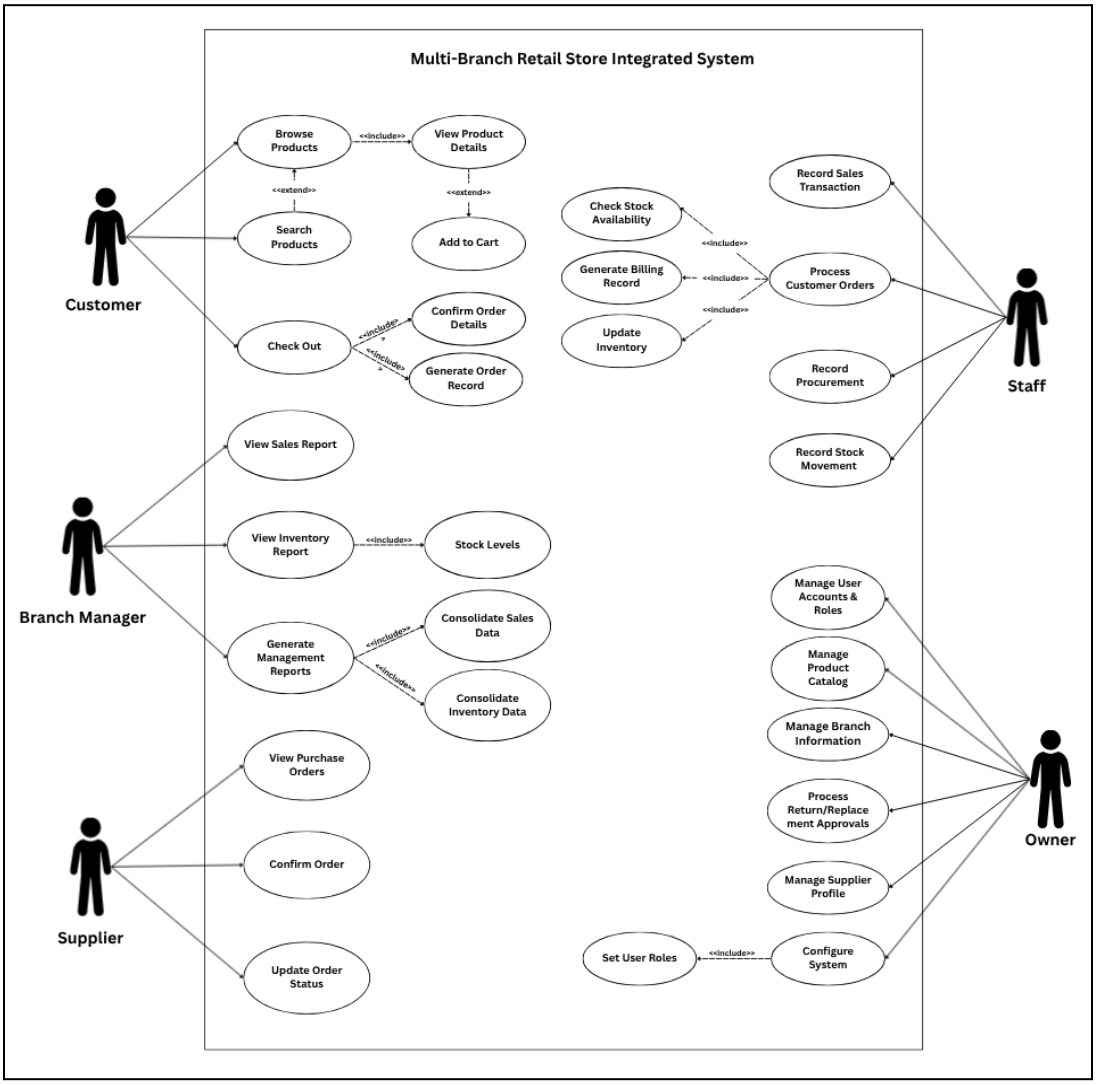
Deliveries are recorded by Staff in the system, ensuring that inventory records are updated in real time. This improved workflow enhances coordination, reduces manual intervention, and ensures that all branches operate using consistent and up-to-date information.

Use Case Diagram

The Use Case Diagram identifies the actors involved in the system and their interactions with its functionalities.

Figure 7

Use Case Diagram



The Customer interacts with the system by placing orders and receiving updates regarding order status and confirmations. The Staff handles transaction processing, inventory updates, and recording of sales and deliveries within the system.

The Branch Manager monitors branch operations by accessing sales and inventory reports, as well as reviewing stock levels and procurement requests. The Owner oversees

overall business operations by managing products, approving procurement, and reviewing consolidated reports across all branches.

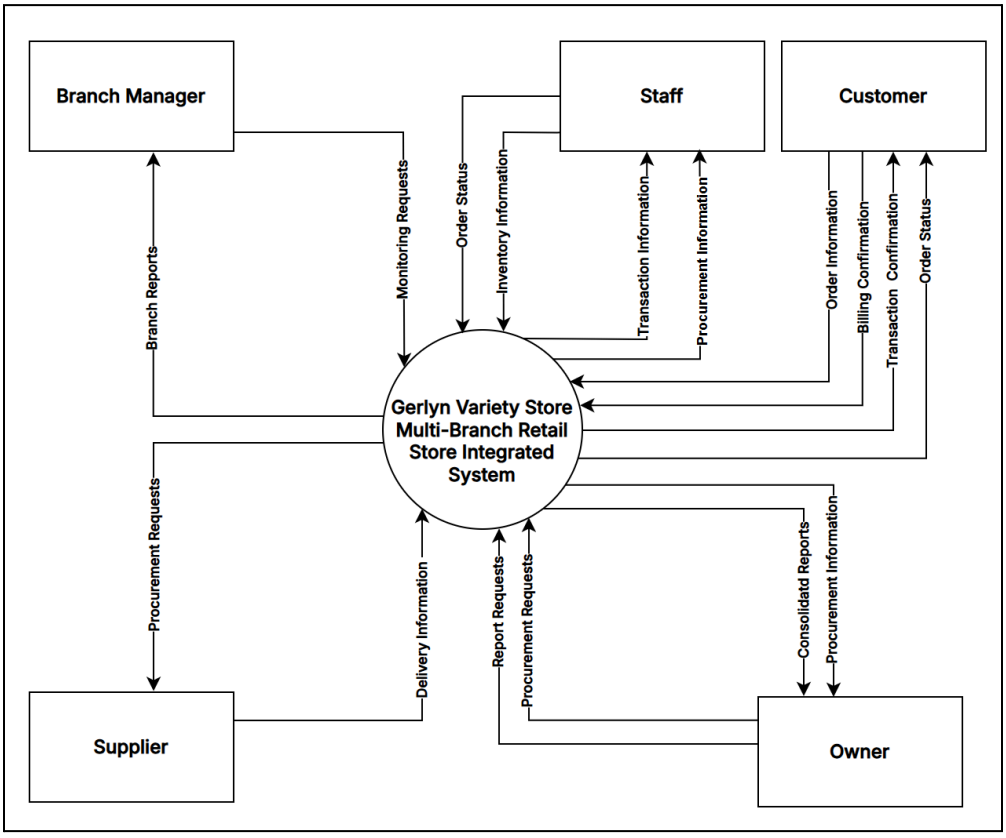
The Supplier participates in the procurement process by receiving purchase orders and providing delivery confirmations. The diagram illustrates how each actor interacts with the system to support coordinated and efficient business operations.

Context Diagram

The Context Diagram shows a high-level overview of the system's scope that identifies the external entities that interact with the integrated solution and the flow of information in Gerlyn Variety Store.

Figure 8

Context Diagram



The diagram shows that the Customer provides order information and receives order confirmation, billing confirmation, transaction confirmation, and order status from the system. The Staff submits transaction information and procurement information and receives inventory information and order status from the system to support daily operational activities. The Branch Manager sends monitoring requests to the system and receives branch reports in return. The Supplier provides delivery information to the system and receives procurement requests to support stock replenishment coordination. Meanwhile, the Owner sends report requests and procurement requests to the system and receives consolidated reports and procurement information in return.

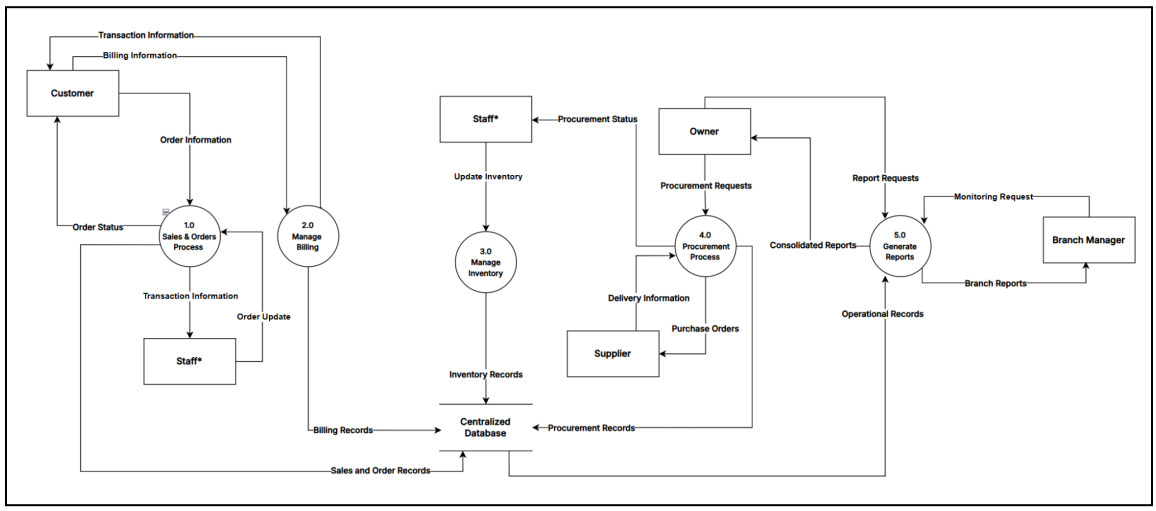
The Context Diagram shows how external entities exchange information with the system to support operational efficiency, coordination, and managerial decision-making across the multiple branches of Gerlyn Variety Store.

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates the logical movement of information within the system, tracing how data is processed, stored, and transformed across different business levels in Gerlyn Variety Store. Through this visualization, the system ensures that data remains consistent and accessible across all branches which eliminates the information silos present in the current manual process.

Figure 9

Data Flow Diagram



This diagram shows how major information flows move through the core processes of the system. The diagram breaks down the proposed system into five main processes, namely 1.0 Sales & Orders Process, 2.0 Manage Billing, 3.0 Manage Inventory, 4.0 Procurement Process, and 5.0 Generate Reports, with the Centralized Database serving as the main repository of operational records.

Process 1.0 handles order information from the Customer and transaction-related coordination with Staff, then provides order status to the Customer while storing sales and order records in the Centralized Database. Process 2.0 manages billing information associated with customer transactions and produces transaction information and billing records that are recorded in the Centralized Database. Process 3.0 handles inventory updates from Staff and maintains inventory records in the Centralized Database to support stock monitoring and inventory control. Process 4.0 manages procurement requests from the Owner and delivery information from the Supplier, then generates purchase orders for the Supplier, provides procurement status to Staff, and stores

procurement records in the Centralized Database. Process 5.0 uses report requests from the Owner, monitoring requests from the Branch Manager, and operational records from the Centralized Database to generate branch reports for the Branch Manager and consolidated reports for the Owner.

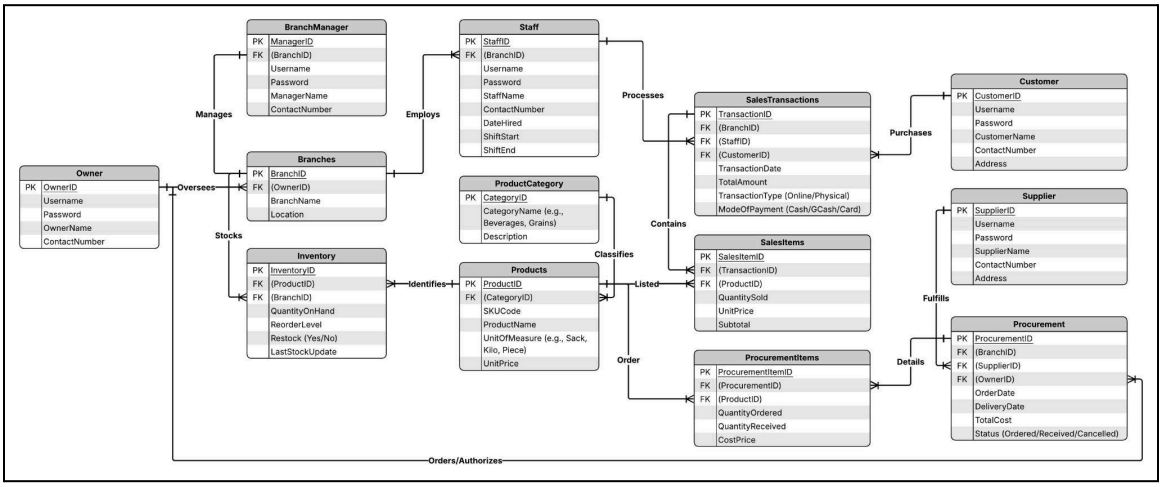
The DFD Level 0 presents a simplified and structured view of how the proposed system processes, stores, and transforms operational information to support sales and order processing, billing management, inventory control, procurement coordination, and report generation across the business operation.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) represents the logical architecture of the Gerlyn Variety Store Multi-Branch Retail Store Integrated System. It serves as a visual blueprint that illustrates how data entities are structured and how they interact to support the business' multi-branch operations. Through implementing a relational model, the system replaces the fragmented, paper-based records of the current manual process with a synchronized environment where processes are logically linked.

Figure 10

Entity Relationship Diagram



The entities of the system are structured to ensure that all business data can be cross-referenced and verified across multiple locations and sales channels. The Branches entity acts as the central hub connecting location-specific data, including the BranchManager overseeing operations, the Staff assigned to the counter, and the Inventory levels. The Owner entity maintains a high-level relationship, overseeing all Branches and serving as the sole authority for Procurement to ensure centralized financial control over all acquisitions.

The Staff entity is connected directly to SalesTransactions to ensure every sale is traceable to the specific employee who processed it. This structure enforces Segregation of Duties: Staff handle transactions, while the BranchManager monitors inventory and updates the Restock (Yes/No) status to flag items for the Owner's review. To adapt to modern retail needs, the SalesTransactions entity distinguishes between "Online" and "Physical" transaction types that allows the system to track orders from different channels while deducting from the same unified Branch Inventory.

The Products entity serves as the master reference that utilizes a unique SKU (Stock Keeping Unit) Code and a Standardized Unit Price. This ensures that product identities and costs remain consistent whether an item is being purchased by the Owner or sold by the Staff to eliminate pricing discrepancies in multi-branch reporting. On the supply side, the Supplier entity fulfills Procurement orders initiated by the Owner. These orders are broken down into ProcurementItems, which systematically update the Inventory once received. This integrated flow ensures that every data point, from the Owner's initial purchase to the final customer sale, is captured to establish a reliable digital audit trail.

Business Rules

The following business rules define the logical relationships between the system's entities, specifically focusing on how Primary Keys (PK) connect to Foreign Keys (FK) to maintain data integrity:

1. An owner oversees many branches to maintain centralized control over multiple store locations.
2. An owner authorizes many procurement transactions to ensure all spending and stock actions are personally approved.
3. A branch manages many inventory records to track the specific stock levels available at that physical location.
4. A branch employs many staff members to handle daily sales operations under a single BranchID.

5. A branch manager monitors many inventory levels to identify which products have hit a low-stock level.
6. A staff member processes many sales transactions to ensure every customer purchase is linked to a specific employee for accountability.
7. A staff member verifies many procurement receipts to confirm that the physical goods delivered match the digital order.
8. A product identifies many inventory records using a unique SKU Code to ensure the same item is recognized across all branches.
9. An inventory record tracks a specific product's quantity to prevent over-selling and provide real-time stock visibility.
10. An inventory record triggers a restock status when the quantity on hand falls below the pre-defined reorder level.
11. A sales transaction contains many sales items to allow the system to track multiple products within a single customer purchase.
12. A sales item lists the specific quantity sold for a single product within a larger customer transaction.
13. A procurement transaction updates branch inventory records only after a formal receipt validation to ensure physical stock matches digital records.
14. A supplier fulfills many procurement transactions.

Tools and Technologies

The system will be developed using appropriate development tools and database management systems. Modeling tools will be used to create workflow and system diagrams. The selection of tools will be based on feasibility and sustainability.

Development Platform

The proposed system will be developed using a web-based development platform. Web technologies such as HTML, CSS, and JavaScript will be used to design the system interface, while Python with the Django framework will be used for system development.

A centralized web-based platform is chosen because it allows all branches of the enterprise to access and share synchronized operational data through a unified system. Transactions performed in one branch are immediately reflected across the system, supporting real-time information flow and coordination. Supporting real-time information flow and coordination. This approach eliminates data silos and ensures that users from different branches operate using the same updated information. The web-based nature of the system allows access through standard browsers without requiring complex installation, making it practical and scalable for multi-branch operations

Database System

The system will utilize MySQL as its database management system. MySQL is a relational database system capable of storing, organizing, and managing structured data such as product records, sales transactions, inventory information, supplier details, and procurement documentation.

MySQL is selected because of its reliability in handling large volumes of data, and compatibility with web-based applications. It also supports centralized data storage, allowing information from multiple branches to be consolidated into a single database. This enables managers to monitor sales, track inventory levels, and generate reports based on real-time operational data.

Modeling Tools

Modeling tools will be used to design and visualize the structure and processes of the proposed information system. Diagramming tools such as Draw.io or similar modeling software will be utilized to create system diagrams including Use Case Diagrams, Data Flow Diagrams, Activity Diagrams, and Entity Relationship Diagrams.

These diagrams help illustrate how users interact with the system, how data moves through different processes, and how system components are organized. The use of modeling tools ensures that the system design is clearly documented and easily understood by developers, stakeholders, and evaluators during the development and evaluation of the project.

Project Timeline

The project will follow a structured timeline to ensure systematic completion of all phases.

Table 1

Gantt Chart

GANTT CHART												
TASK	DEC 2025	JAN 2026	FEB 2026	MAR 2026	APR 2026	MAY 2026	JUN 2026	JUL 2026	AUG 2026	SEP 2026	OCT 2026	NOV 2026
Topic Approval												
Chapter 1-3 Drafting												
Proposal Defense Preparation												
Proposal Defense												
Process Analysis & Data Gathering												
System Requirements Documentation												
System Design												
System Development												
System Testing & Refinement												
User Evaluation & Data Analysis												
Final Documentation Writing (Chapters 4-5)												
Final Defense												

The figure displayed the project timeline, highlighting planning, data gathering, architectural design, development phases, and testing. Each task was aligned with specific months to ensure systematic progress and timely completion.

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